

Markscheme

Mathematics: analysis and approaches
Practice question bank

83. (a)

\$5000 is invested at a nominal annual interest rate of 4%, compounded quarterly.

Find the value of the investment after 3 years, and the effective annual interest rate, correct to 3 significant figures.

$$FV = 5000 \left(1 + \frac{0.04}{4} \right)^{12} \quad (M1)$$

$$FV \approx \$5634.13 \quad \text{A1}$$

$$\text{effective annual rate: } (1.01)^4 - 1 \quad (M1)$$

$$\approx 4.06\% \quad \text{A1}$$

[6 marks]